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Agribusiness Supply Chains: From Green to Regenerative

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ABSTRACT

Supply chain management has traditionally prioritized maximizing profits, creating persistent tensions with social-ecological systems (SES) despite sustainability efforts. Regeneration offers an alternative, ecocentric approach, yet it remains unclear how supply chains can operate within SES boundaries. This research examines how focal firms promote regenerative supply chains (RgSCs) through a multicase analysis of three Latin American agribusinesses. Findings show that these firms leverage their coordination roles to develop two distinctive capabilities: mobilizing deep ecological knowledge and reconfiguring market engagement. Through routines of acquiring, applying, codifying, and exchanging ecological knowledge, firms transform it into a strategic asset. In parallel, by reframing value, shaping biodiverse offerings, and calibrating supply and demand, firms reconfigure their engagement with consumers. The recursive alignment between these two capabilities enables firms to reconnect production and consumption with the dynamics of living systems through continuous adaptation. The study contributes to theory by empirically unpacking the shift from harm reduction to regeneration, developing a model of RgSC capability formation, and introducing the concept of recursive alignment. Building on this contribution, we also reinterpret the Triple-A framework of agility, adaptability, and alignment through a regenerative lens. In RgSCs, agility reflects synchronicity with ecological processes, rather than speed; adaptability emerges as a shared relational capability; and alignment becomes a recursive process integrating ecological and market feedback loops to sustain regeneration. For practitioners, advancing RgSCs requires redefining success beyond financial metrics to include ecosystem health and community wellbeing and shaping demand around ecological boundaries through new forms of collaboration, incentives, and education.

1 | Introduction

Given their global reach and crucial role in shaping production and consumption, supply chains (SC) are “at the heart of societal challenges” (Touboul et al. 2020, 36) and crucial to transformative change. Conventional supply chain management (SCM) has long relied on “paradigmatic assumptions” (Gualandris et al. 2024) that prioritize profit maximization via continuous growth, economies of scale, and the efficient use of natural and human capital as production inputs (Friedman 1951). Performance is typically driven by top-down mechanisms, aiming to meet demand at the lowest possible cost (Fisher 1997; Gereffi 1994; Porter 1989). This efficiency focus often overlooks

the broader SES where supply chains operate. Despite progress in sustainable SCM (Seuring et al. 2022), the field remains focused on reducing environmental and social harm, through material reuse and pollution control (Ellen MacArthur Foundation, n.d.), rather than creating positive, regenerative outcomes (Pagell and Shevchenko 2014) that would enhance SC's potential for transformative change.

Only recently has there been interest in RgSCs that seek to incorporate strong sustainability principles into supply chain research and practice (Gualandris et al. 2024; Luzzini et al. 2024). RgSCs are defined as “inter-organizational networks that sense and embrace surrounding living systems,

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aligning their decision-making and actions to these systems' structures and dynamics, in a way that allows for such systems to gain strength, build resilience, and sustain life" (Muñoz and Branzei 2021, 510). Their design, coordination, and governance follow the principles of proportionality, reciprocity, and poly-rhythmicity (Gualandris et al. 2024). Proportionality aligns supply chain activities with the biological boundaries and social capacities of local ecosystems and communities. Reciprocity emphasizes mutual benefit and interdependence between supply chains, local communities, and ecosystems. Poly-rhythmicity underscores the importance of acknowledging and working within the multiple, overlapping cycles of SES. The main challenge is bridging these theoretical principles with actual company practices to introduce regeneration across supply chains.

Through an exploratory, multicase analysis of three regenerative agribusinesses in Latin America, this research attempts to address two questions: What capabilities do focal firms develop to promote regenerative supply chains? What processes or routines do they enact that show the deployment of those capabilities? Capabilities, as described in the resource-based and dynamic capabilities literature (Barney 1991; Teece et al. 1997; Teece 2007), provide firms with the ability to create value and continuously reconfigure their resources in response to shifting environmental and market conditions. Capabilities are often thought of as high-level routines or bundles of routines that are learnt, highly patterned, and repetitive (Winter 2003; Peng et al. 2008).

The cases included here, although drawn from the same industry and regional context, exhibit diverse business models and market orientations, allowing for a rich comparison of how regeneration is advanced across supply chains. This research shows that focal firms cultivate two distinctive capabilities that are critical to promoting RgSCs. First, they purposefully mobilize deep ecological knowledge of their specific geographic and social contexts to align supply chain operations with local ecosystems. Second, they reconfigure their market engagement by educating clients and consumers on the spatial and temporal dimensions of ecological processes that underpin regenerative practices. This awareness reframes expectations around product availability, while emphasizing enhanced quality and biodiversity.

The two capabilities appear through distinct but connected routines. Firms mobilize ecological knowledge by acquiring, applying, codifying, and exchanging localized ecological knowledge within their business models. They also reconfigure market engagement through processes that include shaping biodiverse offerings, reframing value, and calibrating demand and supply. Importantly, these two capabilities do not operate independently as ecological knowledge continuously informs market engagement; in turn, interactions with markets shape how ecological knowledge is mobilized upstream. We refer to this ongoing, bidirectional process as recursive alignment.

This study examines the shift from harm reduction to regeneration. It contributes to theory by (1) presenting a model of RgSC capability development that reconnects production and

consumption with living systems; (2) introducing the concept of recursive alignment between these capabilities, showing that regeneration is a process of continual adaptation, not a fixed end state; and (3) extending the literature on capabilities in supply chain management with regenerative interpretations of agility, adaptability, and alignment, which challenge longstanding anthropocentric assumptions in supply chain design, coordination, and governance.

For practitioners and policymakers, our findings highlight that advancing RgSCs requires rethinking both performance metrics and market logics. Organizations must evaluate success through the wellbeing of the SES that sustain production—embedding indicators such as soil health, biodiversity metrics, and livelihoods—along financial metrics. Managing demand also shifts from responding to markets to shaping demand around ecological boundaries and cycles, aligning consumption with what ecosystems can sustainably provide over time. These lessons call for new forms of collaboration among firms, policymakers, and educators to embed regenerative principles in business practice, support systems, and professional training.

2 | Conceptual Framework

2.1 | Sustainability in Supply Chains

SCM has evolved over the last few decades, shifting from a focus on cost savings and operational efficiencies (Pagell and Shevchenko 2014) to the recent emergence of RgSCs (Muñoz and Branzei 2021). Although this paper does not review this evolution systematically, it highlights key elements that differentiate RgSCs from earlier sustainability conceptualizations. We offer a model to better understand current developments in practice—among companies, cooperatives, and entrepreneurs involved in the production, distribution, and sale of products—whose strategies depart substantially from previous supply chain models that attempted to address social and environmental impacts.

2.1.1 | Greening the Supply Chain: From Compliance to Value Creation

Early approaches in the 1990s were largely grounded in calls to “green” the supply chain by focusing primarily on compliance with environmental and social regulations (Srivastava 2007). Later in the decade, approaches to greening evolved from reactive to proactive by innovating across operations to encompass remanufacturing, recycling, reverse logistics, and waste management (Srivastava 2007). Circular economy (CE) thinking introduced new principles and tools for supply chains sustainability, such as life-cycle assessment and material flow analysis (Suárez-Eiroa et al. 2019; Ellen MacArthur Foundation, n.d.). By focusing on energy and material flows, CE seeks to increase material efficiency and extend producers' responsibility beyond sale. This fosters supply chain networks that close material loops between suppliers and consumers. CE offers an integrated sustainability framework that addresses environmental, economic, and, at times, social goals (Zhang et al. 2021). Life cycle assessments

evaluate environmental impacts by tracing energy and material flows (Eskandarpour et al. 2015). CE-based supply chains focus on identifying win-win opportunities where reducing environmental impacts lead to value recovery and increased profitability (Amir et al. 2023, 2684).

Pagell and Shevchenko (2014) recognize that sustainability has evolved from a *fringe* concern in the 1990s to a mainstream subject within SCM. Despite this progress, they argue that the field still lacks the conceptual and methodological depth to achieve *truly sustainable supply chains*, because much of the research prioritizes incremental improvements and win-win situations, ignoring trade-offs, radical innovation, and forward-looking methods. These approaches have also been criticized for subordinating environmental and social goals to economic interests (Montabon et al. 2016), rendering investments in sustainability contingent on their economic profitability.

2.1.2 | Supply Chains as Embedded Socioecological Systems

Considering accelerating ecological degradation and approaching planetary tipping points, scholars now call for a re-orientation of SCM toward an ecological foundation, placing environmental systems at the core of all organizational activity (Hellweg et al. 2023). This ecological grounding emphasizes the intrinsic value of ecosystems, advocating for their protection and restoration as preconditions for both social and economic well-being.

The shift toward understanding supply chains as embedded within SES represents a critical departure from earlier sustainability paradigms. This perspective recognizes that supply chains are not isolated economic entities but integral components of living systems, subject to feedback loops, constraints, and interdependencies (Wieland 2021). The discipline of SCM, Wieland (2021) argues, has been ill-equipped to effectively respond to complex challenges and rapidly evolving conditions, mainly due to two implicit assumptions. First, SCM theories often assume the long-term stability of key conditions, such as the continuous availability of natural resources for manufacturing. This assumption, he argues, “reduces the system’s ability to adapt when these conditions change” (59). Second, there is a tendency to presume that “a supply chain can be isolated from the rest of the world” (60), thereby ignoring its embeddedness within broader ecological, social, and economic systems.

To address these limitations, Wieland (2021, 60) proposes replacing the traditional engineering-based vision of supply chains with a new focus on “change, unpredictability, persistence, and transformability”. He advocates for reinterpreting supply chains through an ecological lens, grounded in the concept of panarchy (Gunderson and Holling 2002), which offers a framework for understanding how systems adapt over time, space, and meaning. This perspective conceptualizes supply chains as SES, acknowledging the complex interdependencies between human and natural systems and the need to understand how supply chains persist, adapt, or transform in the face of environmental challenges.

2.1.3 | Regeneration: Restoring and Ensuring the Resilience of Ecosystems

Regeneration represents a paradigmatic shift in supply chain thought. RgSCs reframe organizational purpose from minimizing harm to actively restoring ecological and social systems, in response to complex challenges and rapidly evolving conditions, as those posed by climate change and biodiversity loss. According to Muñoz and Branzei (2021, 510), RgSCs are “inter-organizational networks that sense and embrace surrounding living systems, aligning their decision-making and actions to these systems’ structures and dynamics, in a way that allows for such systems to gain strength, build resilience, and sustain life.”

This ecocentric orientation acknowledges the biophysical boundaries within which all supply chain activities occur and situates human enterprise as a regenerative force within larger ecological contexts. The regenerative paradigm emphasizes coevolution, resilience, and reciprocity between supply chains and natural systems. By integrating insights from systems theory and ecological economics, it challenges the anthropocentric orientation of conventional sustainability frameworks and instead promotes an ontology of interdependence (Hahn and Tampe 2021; Heikkurinen et al. 2016).

In this view, communities are repositioned from being mere providers of raw materials to active participants in regenerative processes. Montabon et al. (2016) argue that community well-being is placed at the core of supply chain strategies, thereby redefining success in terms of resilience and life-sustaining outcomes for nature and people and recognizing the value of indigenous and local communities’ knowledge grounded in their longstanding relationships with the land.

Based on empirical evidence and synthesizing emerging arguments from a variety of disciplines, Gualandris et al. (2024) propose three principles that govern RgSCs with implications for supply chain coordination, governance, and resilience. These three principles are proportionality, reciprocity, and poly-rhythmicity. Proportionality involves aligning production and consumption within the ecological boundaries of the SES, thereby restoring balance and biodiversity. Reciprocity emphasizes designing supply chain relationships that create mutual benefits among all actors—including nature—whereas poly-rhythmicity reflects the synchronization of operations with the natural temporal cycles and ecological rhythms of the systems in which they are embedded (Gualandris et al. 2024). These principles suggest that firms must develop capabilities not only to adapt to changing market conditions but also to coevolve with the ecosystems that sustain them (Bag et al. 2025). This requires reimagining organizational objectives to prioritize ecological resilience, social well-being, and systemic balance over short-term economic gains.

Despite its transformative vision, regenerative supply chains remain constrained by several practical limitations: Operationalizing proportionality, reciprocity, and poly-rhythmicity within complex global supply networks presents significant governance and measurement challenges. Questions such as how to orchestrate this shift across supply chains, what organizational capabilities are needed to bring to practice these

principles, and what common frameworks to use for evaluating regenerative outcomes remain unexplored.

2.2 | Capabilities and Routines in Supply Chains

In advancing RgSCs, routines and capabilities are key theoretical and practical constructs that bridge every day, microlevel organizational activities with long-term macrolevel adaptive strategies (Felin et al. 2015). The literature on resource-based view emphasizes that competitive advantage derives from unique resources and firm-specific capabilities (Barney 1991), whereas the dynamic capabilities literature stresses that what matters is the ability to continuously sense, seize, and reconfigure resources in response to shifting environmental and market conditions (Teece et al. 1997; Teece 2007). Dynamic capabilities can be understood as processes or macroroutines by which companies identify, acquire, and transfer knowledge (Parmigiani and Howard-Grenville 2011). Under this perspective, routines lead to both continuity (strengthening a company's competitive advantage) and change (by allowing individuals to select from a portfolio of options, recombining routines and developing new ones), as a response to challenges imposed by the environment (Parmigiani and Howard-Grenville 2011).

In supply chains, Lee (2004) identified three capabilities for effective management: agility, adaptability, and alignment (i.e., the AAA capabilities). Agility is defined as responding quickly to short-term market changes; adaptability is the capacity to innovate and transform in response to long-term structural changes, driven by socioeconomic or technological shifts, whereas alignment consists of calibrating and coordinating actions across all links in the chain from extraction to delivery. More recently, Erhun et al. (2021) discuss those AAA characteristics in light of current social and environmental awareness. For this perspective, agile supply chains need to respond quickly to both market changes and pressures from a broader set of stakeholders. Adaptable supply chains innovate to respond to sustainability trends and foster cooperation with suppliers. Finally, aligned supply chains ensure transparency and information sharing among all links to secure the implementation of sustainability principles.

In a similar vein, Sodhi and Tang (2021) provide an interpretation of the Triple-A framework that embeds supply chains within their ecosystem, both natural and institutional, referring to the natural resources the supply chain depends upon and the stakeholders with whom companies interact. The consequences of such inclusion are supply chains that are more responsive to a larger group of stakeholders; more flexible in adapting to environmental and social changes, as well as to market changes; and better coordinated across actors to secure alignment in obtaining economic, social, and environmental goals. Yet, under these models, capabilities emerge as responses to market pressures and are defined as the ability of companies and their suppliers to adapt to these requirements. They are not the result of an intrinsic desire to change how markets operate and how supply chains are organized under an SES logic. Responding to this constraint, emerging research posits that RgSCs demand capabilities oriented

toward life-sustaining outcomes, which require sensing, synchronizing, and cocreating with living systems (Muñoz and Branzei 2021; Gualandris et al. 2024). The challenge lies in identifying and cultivating these capabilities across diverse actors and embedding a regenerative logic into decision-making.

In terms of routines or processes, Pagell and Wu (2009) identify a wide range of sustainable SC applied practices that focus on managing risks, engaging stakeholders, and closing loops. The SC literature speaks colloquially of these processes or routines as practices that aim to minimize negative externalities—such as waste and emissions—through supplier collaboration, eco-design, and improved product life-cycle management. Similarly, Hong et al. (2018) emphasize how sustainable SCM practices contribute to enterprise performance, particularly by improving resource efficiency and reducing dependency on nonrenewable inputs. These approaches contribute to sustainability goals, although largely grounded in a harm-reduction paradigm.

RgSC demands organizational routines and social interactions that align the behaviors of all participants toward resilience, involving continuous learning and the concerted efforts of actors from origin to consumption. As Muñoz and Branzei (2021) and Gualandris et al. (2024) emphasize, regenerative supply chains must intentionally enhance SES health through coordinated multiactor efforts.

The absence of a well-defined set of supply chain routines and capabilities in this literature presents a significant gap in understanding how RgSCs models can be enacted beyond the farm gate. A central challenge, therefore, is to conceptualize how sustainable SCM capabilities might evolve to support regenerative outcomes. Some dynamic capabilities in the sustainable SCM literature—such as stakeholder integration and adaptive learning—may serve as foundational elements for RgSC development. However, additional and distinct capabilities appear necessary to support a shift from sustainability to systemic regeneration. This need is especially pronounced given the place-based characteristics of natural environments, which demand not only adaptability but a reconceptualization of what drives the transformation of supply chains toward regeneration. This research addresses this theoretical gap by examining how focal firms promote RgSCs, revealing the emergence of previously unidentified capabilities essential to this transition.

3 | Methods

3.1 | Research Design

A qualitative case study approach was adopted, given that research on RgSC “will likely be inductive and qualitative to clarify and elaborate core constructs” (Gualandris et al. 2024, 62). Case studies help address the “how” and “why” of a phenomenon (Yin 2015). For theory building, we drew on established grounded-theory procedures to systematically move from empirical material to conceptual abstraction, with Gioia et al. (2013) serving as an operational guide alongside classic grounded-theory foundations (e.g., Strauss and Corbin 1994; Glaser and Strauss 2017).

3.2 | Case Selection and Sample Description

Case selection used purposeful sampling (Creswell 2013) to identify “information-rich cases” (Patton 2002). Firms met three criteria: (i) They actively regenerated or prevented further degradation of SES; (ii) their business models were innovative and different from conventional agribusiness logics; and (iii) they operated in Latin America. The region offers a compelling context for studying RgSCs due to its rich biodiversity, the prevalence of extractive agricultural practices, and the emergence of regenerative alternatives. Although intraregional diversity exists, the region shares contextual features—such as land tenure insecurity, regulatory inconsistencies, and market pressures—that shape its agribusiness dynamics.

From an initial pool of regenerative organizations documented in prior projects, we selected three focal firms—anonimized as C1, C2, and C3 (see Table 1). This ensured analytical depth and narrative coherence. Focusing on three cases streamlined the storyline and provided sufficient space to develop detailed accounts of regenerative supply chain routines. Sufficient diversity was retained to capture meaningful variation across organizational contexts. Other cases from prior projects were not included in this analysis but provided supporting evidence, helping to strengthen and contextualize the arguments from the three focal cases.

The selected three firms were distinctive because of their ability to drive regeneration across entire supply chains. They influenced upstream suppliers and downstream customers. This ability to promote regenerative practices beyond their own boundaries made them ideal for understanding RgSCs in practice. The firms also differ across several business dimensions. Some have more control over downstream operations, whereas others focus more on upstream activities. They also vary in market orientation, targeting either national or international clients. Despite these differences, the process of developing RgSc capabilities was similar. Thus, these firms served as general cases. To develop the comparative logic, the process through which each firm builds regenerative capabilities was treated as the embedded unit of analysis for data collection, interpretation, and model construction to address the research questions.

3.3 | Data Collection and Analysis

Data collection followed a unified case-study protocol informed by prior literature and expert consultation (Stuart et al. 2002; Yin 2015). The protocol required site visits and semistructured interviews with founders, managers, employees, and farmers. For consistency, other secondary sources were reviewed to contextualize and triangulate findings. All data collection activities, carried out by a group of researchers, were recorded and transcribed and subjected to triangulation to ensure reliability and validity (Yin 2015). Appendix A lists all data sources, interviewees, their roles, and relevance.

The case-study protocol¹ served as initial guidance for framing the research questions on RgSCs. Rather than relying solely on

case reports, the research drew on the original interview recordings and transcripts, which serve as the primary data for the coding. To ensure tracing across time, additional interviews were conducted in 2025. Furthermore, authors conducted additional secondary data analysis based on information provided by interviewees to capture developments since the original data collection.

The authors analyzed data using an open-ended, theory-building approach that evolved iteratively during data collection (Strauss and Corbin 1998; Gioia et al. 2013). This approach ensured methodological rigor by providing a systematic framework for coding and analysis (Magnani and Gioia 2023). All transcripts and secondary materials were coded inductively to generate first-order codes, using the richly detailed cases to construct the initial structure, and the less detailed cases to refine it (Maxwell 2013). Throughout, insights from SCM literature were iteratively compared with emerging codes to ensure alignment with and contribution to existing knowledge (Gioia et al. 2013). Second-order themes were independently reviewed by two of the researchers and finalized through collaborative discussions. Aggregate dimensions, abstracted from these themes, were developed in a similar way, yielding a cohesive framework to address the research question. Figure 1 shows the data structure.

Next, an illustrative grounded model was developed to capture key concepts, themes, and dimensions, along with their interrelations (Magnani and Gioia 2023). In the findings section, results of this model are presented. Aggregate dimensions and second-order themes were distilled into a coherent framework, highlighting their interrelationships. To strengthen confidence in these interpretations, authors sought feedback from key informants and engaged in peer debriefing with colleagues not involved in the study (Gioia et al. 2013). Each dimension is presented individually, providing its definition and significance and developing supporting arguments. To ensure rigor, illustrative data excerpts were included, such as interview quotes, alongside the analysis.

4 | Findings

Data reveal that focal firms promoting RgSCs develop two distinctive capabilities, depicted in the model in Figure 2. First, they build the capacity to mobilize deep ecological knowledge. Second, they develop the capability to reconfigure market engagement. Taken together, these capabilities form a recursive pattern in which each one informs the other. Below, we present each capability and the set of routines through which they are deployed.

4.1 | Mobilizing Ecological Knowledge

This capability develops primarily upstream, though it is continually informed by interactions with markets. It refers to the ability to access, adapt, codify, and exchange diverse ecological knowledge—including scientific, experiential, indigenous, and local knowledge (as defined by IPBES 2019)—to guide action across the supply chain. Ecological knowledge is the

TABLE 1 | Sample details.

C1				C2		C3	
Founding year	2006 (regenerative in 2015)			2016 (operations in 2009)		2010 (operations in 2004)	
Type of organization	Family-owned company; for-profit company			For-profit company		Not-for-profit association (263 members)	
Location	Guatemala (dry corridor)			Brazil (the Amazon)		Ecuador (the Amazon)	
Employees	14 (including the 2 owners)			23 (including 2 managers)		23 direct employees (including 5 managers), 80 seasonal workers for value-adding activities.	
Sales	170,000 USD (2022)			30,000 USD (2023)		800,000 USD (2022)	
Impacted territory (ha)	24			40 (34 regenerated, 4 productive areas and 2 infrastructure)		7000	
Main Products	Dairy products, chickens, and eggs			Freeze-dried and pulverized nonconventional food plants (in Portuguese PANCs) Açaí, jambu, ora-pro-nobis, chicória, caruru, taioba, and vinagrera		Toasted cocoa grains, peeled plantain, peeled cassava, dried leaf guayusa, and chocolate bars	
Certifications	No certifications			Organic Certification (USA, EU, and Brazil) and HALAL HACCP, GFSI, and KOSHER (in process)		Organic Certification (USA, EU, and ECU) EU deforestation-free (In process)	
Position in supply chain	Produces, transforms, autoconsumes and sells directly to consumers. Controls the whole supply chain (B2C)			Produces, collects raw material (from over 75 suppliers), transforms and sells directly to consumer, to health stores and transformers of raw materials (B2C, B2B2C)		Produces and collects raw material (from over 800 suppliers). Performs low-value adding activities and sells to national and international clients (B2B)	
Markets	Local (directly to high-end consumers of Guatemala)			Local (directly to high-end consumers in Brazil) International (directly to high-end consumers and transformers of raw materials in North America, Europe, and Asia)		Local (for plantain and cassava, to transformers of raw materials) International (for cocoa and guayusa, to transformers of raw materials in Asia, North America, and Europe)	
Regenerative aspects	Regeneration as a strategy to access markets and later on as a philosophy Holistic planned grazing, polyculture, edible forests			Regeneration as a strategy for development Syntropic agriculture (Götsch)		Regeneration as a philosophy Chakra system (indigenous ancestral way of living in harmony with the land). Regenerate and preserve Kichwa culture	

Source: Created by the authors.

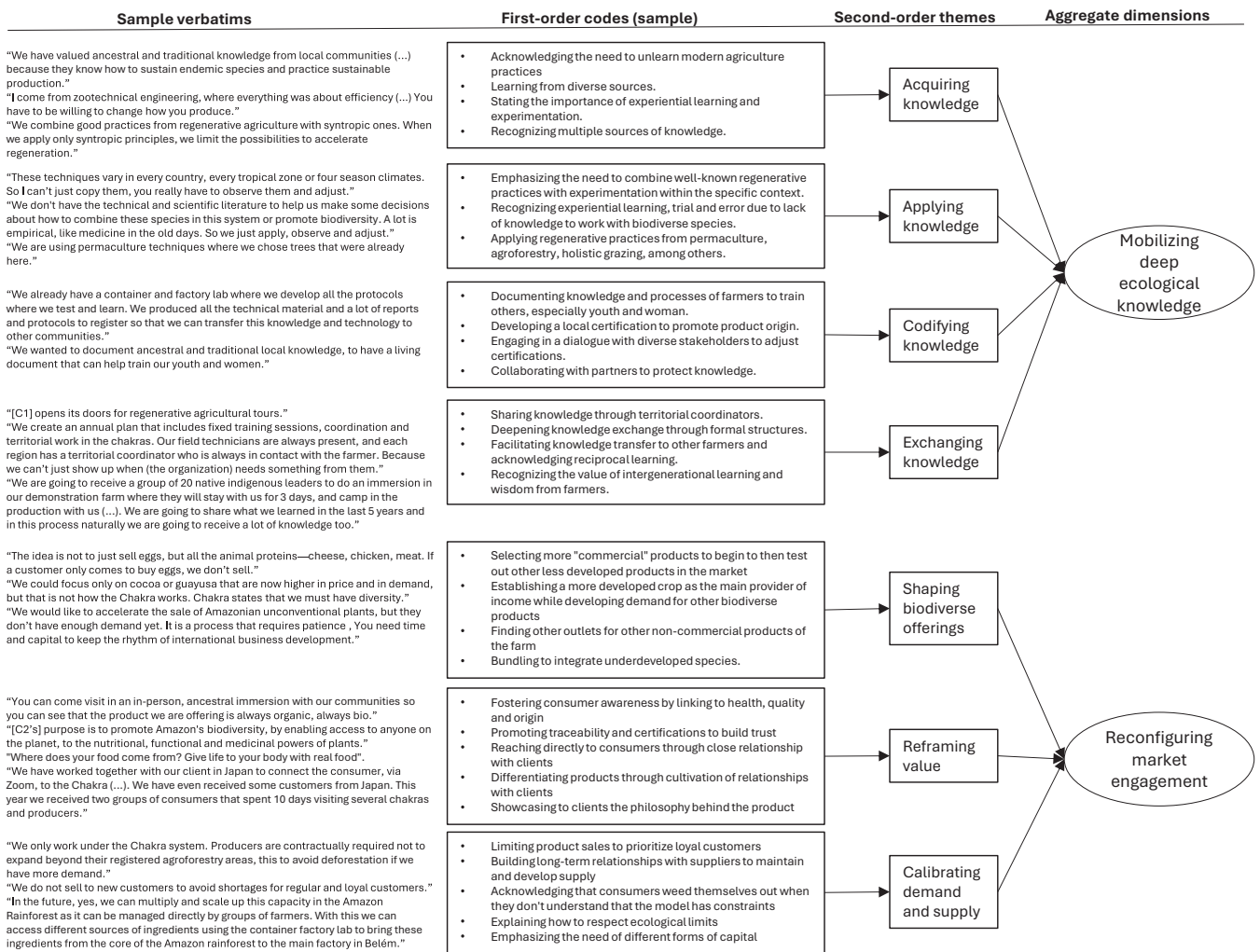


FIGURE 1 | Data structure.

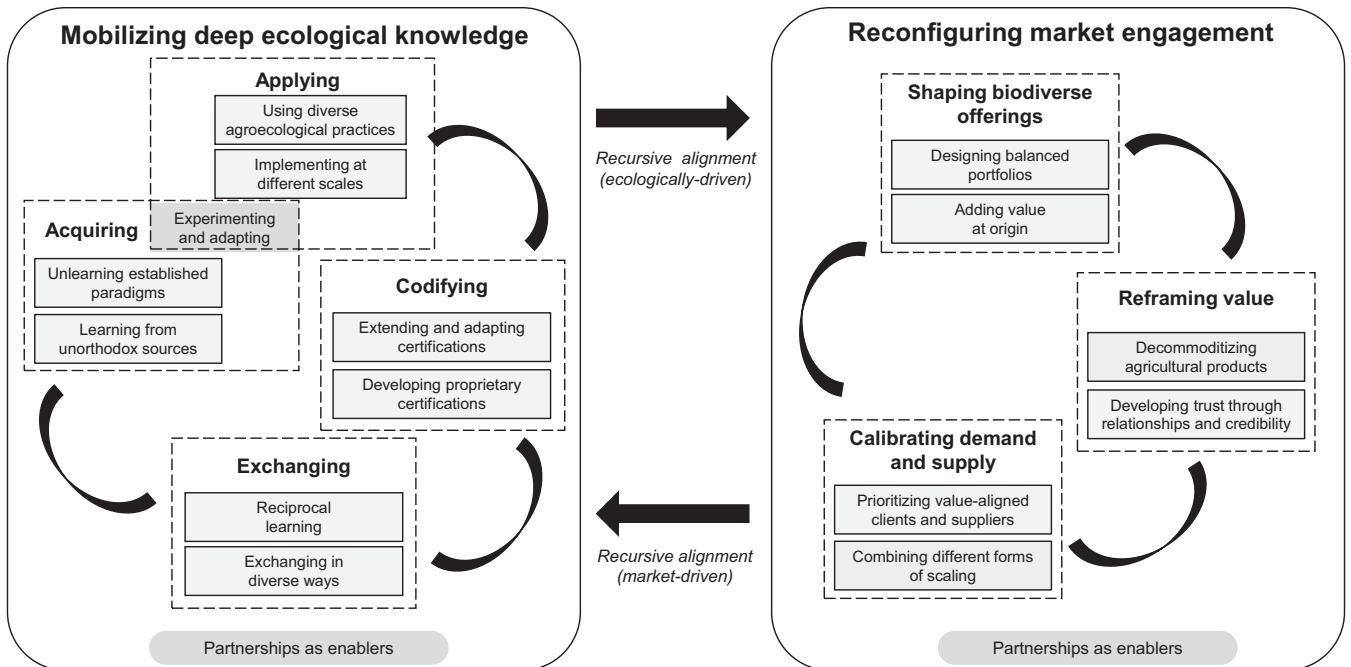


FIGURE 2 | A model of capabilities to promote regenerative supply chains.

system-level understanding of the structures, rhythms, and interdependencies within a specific SES. This capability is inherently place based and relational; without it, sustainability efforts risk becoming top-down, misaligned with local conditions, or unable to scale within socioecological boundaries. This capability emerges through recursive loops of acquiring, applying, codifying and exchanging, which we detail in the following sections.

4.1.1 | Acquiring Knowledge

Acquiring knowledge refers to the routines through which firms actively gather and accumulate relevant ecological knowledge from diverse sources. This place-based knowledge is not readily accessible, often absent from formal agronomy training, and its acquisition requires a deliberate effort. This process entails unlearning conventional assumptions and gathering new knowledge while engaging in continual experimentation. Because ecological knowledge must be interpreted in situ, acquiring knowledge frequently overlaps with applying it.

Unlearning established paradigms is often the starting point. This is usually triggered by some form of disruption, such as declining productivity, rising pollution, or visible ecological degradation. These issues prompt firms to let go of industrial farming approaches rooted in efficiency and control. “I come from zootechnical engineering, where everything was about efficiency (...) You have to be willing to change how you produce,” noted the cofounder of C1. As firms confront these disruptions, they gradually adopt a more holistic perspective that values biodiversity, interdependence and complexity. “We must learn from the limits and wisdom nature imposes” continued the cofounder of C1. Across cases, this perspective also carried cultural meaning, emphasizing land stewardship and the protection of culture: “We are protecting biodiversity, the environment, the forest and our traditional customs” (codirector, C3).

As firms unlearn established logics, they turn to unorthodox sources of knowledge rarely provided by formal institutions. Although some actors have agronomic training, much of the learning occurs from a combination of knowledge held by forward-looking institutions, pioneers in the field, elders (indigenous people), and through experimentation. In C1, for example, the cofounder's son is an agronomist trained at EARTH University, an ecologically oriented agricultural school in Costa Rica, whereas she herself, despite lacking formal education, received specialized training from a renowned agronomist known for sustainable practices. In C2, the founder drew direct inspiration from Ernst Götsch—pioneer of syntropic agriculture² in Brazil. Still, formal training alone is insufficient. As the codirector of C3 noted: “We value ancestral and traditional knowledge from local communities (...) because they know how to sustain endemic species and practice sustainable production.”

Acquiring knowledge is thus inherently tied to experimenting and adapting locally. Learning unfolds through hands-on trials that reveal how regenerative practices interact with soils,

climates, and ecosystems. As the cofounder of C2 explained, “Our team is mostly empirical (...), there's no guidebook.”

This experiential learning process is inherently adaptive and context specific. Firms draw from multiple sources, but these inputs only become meaningful through repeated local trials. The cofounder of C1 emphasized: “You cannot simply copy them [regenerative practices]. It is necessary to observe, to be in your land, to adapt the practices to the place where you work. We go back to being farmers—observing, seeing what fertile soil means here.” Experimentation therefore serves as the mechanism through which regenerative principles are translated into locally viable practices.

Lack of precise measurements further embeds experimentation into the learning process as a primary mode of knowledge acquisition. In the absence of formal tools to quantify soil conditions or microbiological processes, firms rely on iterative trial and error to interpret ecological feedback. As the cofounder of C1 observed, “How can we know how much moisture or microbiology is in the soil? We don't have the resources to measure that. We just know we're producing more and better.” Learning, thus, proceeds through experimental adjustments whose outcomes are assessed through observable changes in productivity and ecosystem health, rather than through formal indicators.

As a result, acquiring knowledge through experimentation becomes difficult to separate from applying in practice. The same iterative cycles that generate ecological understanding simultaneously enact it, blurring the boundary between learning and doing. As the cofounder of C2 summarized; “We just apply, observe and adjust.”

4.1.2 | Applying Knowledge

Applying knowledge refers to the routines through which firms translate ecological knowledge into concrete, on-site actions, across different geographical scales. Like acquiring, applying is a recursive process, unfolding through cycles of testing, observing, and adjusting.

A central aspect of applying knowledge involves using diverse agroecological practices. Our firms drew on known approaches—holistic grazing (C1), edible forest systems (C2), and agroforestry (C3). These approaches were adjusted to local realities. The cofounder of C1 elaborated: “These techniques vary in every country, every tropical zone or four-season climates. So, I can't just copy them.”

Applying extends beyond the firm's own farm to be implemented at different scales. Once practices are refined through local experimentation, firms work with suppliers and neighboring communities to exchange knowledge, adapt it, and apply it in their own contexts. The cofounder of C1 noted: “We maintain ties with neighboring communities (...), we collaborate through training and support for planting and maintenance. Over the last 5 years, we have reforested the area, planting more than 20,000 trees.” At the landscape level, firms work to strengthen connectivity and promote stewardship

beyond their immediate operations. One of the codirectors of C3 explained, “as a community, we have always valued forest connectivity and our own connection with the forest as a whole. Our associates deliberately work with the forest because we cannot live without it.”

4.1.3 | Codifying Knowledge

Codifying knowledge refers to the routines of documenting, systematizing, and formalizing ecological practices to preserve and transfer them across the supply chain. It involves turning informal, experience-based knowledge into transferable protocols without losing its ecological or cultural value. Without codification, knowledge remains fragmented, overly dependent on individuals, and vulnerable to loss.

Codifying also reflects a firm's evolving commitment to ecological stewardship. As knowledge matures through repeated use, codifying attempts to retain and transmit it—allowing field technicians, suppliers, and communities to align their actions with broader regenerative aims. In this sense, codification supports both internal continuity and scaling. To codify, firms draw on several mechanisms: adapting and extending external formal systems of certifications (C2 and C3), creating internal codification systems like in-house protocols (C1), or using proprietary certifications (C3).

In two of our cases, existing third-party certifications served as a basis for their own internal standards. Although the practices of our cases frequently exceeded certification requirements, existing certifications provided a degree of standardization and external legitimacy. The C2 cofounder explained: “All our açai is organic (...). For clients in Europe, sales depend on storytelling and certifications.”

Some firms actively worked to reshape these standards to reflect local realities better. Directors from C3 collaborated with auditors to adapt organic certification procedures to the Amazonian context. One codirector from C3 elaborated, “Initially, we faced several difficulties certifying these farms. Auditors didn't know this type of environment or the habitat (...). They asked for straight boundaries, irrigation systems—things that do not exist here in the Amazon. So, together we created new procedures and best farming practice standards for the Amazon.”

Developing proprietary systems represents a second pathway for codifying ecological knowledge. In C2, codification was carried out entirely in-house, by creating an in-house protocol to support future scaling. The cofounder of C2 elaborated: “We already have a container and factory lab where we develop all the protocols where we test and learn. We produced all the technical material and a lot of reports and protocols for book-keeping so that we can transfer this knowledge and technology to other communities.”

C3 formalized local and indigenous knowledge through their own proprietary standards, the Sello Chakra.³ Created in partnership with regional associations, it ensures product traceability while protecting indigenous knowledge. C3 also worked with

the Ecuadorian government to establish the certificate of origin for guayusa leaves, further institutionalizing local practices and reinforcing territorial identity.

4.1.4 | Exchanging Knowledge

Exchanging knowledge refers to the routines through which ecological knowledge is shared among farmers, clients, and partners. Beyond sharing information, exchanges build trust and enable both codified protocols and tacit experiential insights to become understandable and actionable.

Firms and producers operate in dynamic conditions where sharing experiences refines what is known and how it is used. Exchanging feeds back into the other processes: It challenges what has been acquired, tests what has been applied, and highlights new insights to be codified. Exchanging knowledge is a recursive loop that reinforces and expands the learning process, transforming individual insights into a collective capability.

Reciprocal learning and feedback sharpen and deepen prior knowledge. One C2's cofounder described how they use their demonstrative farm for immersion programs: “We are going to receive a group of 20 native indigenous leaders (...). We are going to share what we learned in the last 5 years and in this process, naturally, we are going to receive a lot of knowledge too.” Exchanges are a two-way process by which firms share knowledge while enriching their own.

Exchanging knowledge occurs by formal and informal mechanisms. These include training workshops, technical visits, certification programs, and farmer-to-farmer interactions. A C3 codirector emphasized how exchanges are integrated into daily operations: “We create an annual plan that includes training sessions, coordination and territorial work in the Chakras. Our field technicians are always present, and each region has a territorial coordinator who is always in contact with the farmer.” In C1, the firm has an open-door policy, “[C1] opens its doors for regenerative agricultural tours,” (C1 social media post) and is working toward a more formal structure as explained by C1's cofounder, “We seek to establish a regenerative agriculture school to deepen this exchange with others.”

4.1.5 | Partnerships Enabling the Mobilization of Deep Ecological Knowledge

Partnerships play a critical enabling role across all stages of mobilizing ecological knowledge. Because many firms lack capital, infrastructure, or technical capacity, partnerships with universities, NGOs, and government programs provided scientific training, external funding, and technical support for adapting practices and professionalizing suppliers. C2 collaborated with the Federal Rural University of the Amazon to strengthen syntropic agroforestry practices, whereas C3 worked with local universities to document ecological knowledge and support certification efforts.

Partnerships also helped protect culturally embedded knowledge from misappropriation; as a codirector of C3 noted, “Our

partners [referring to other associations, a university, and the Ecuadorian Government] are doing these studies (...) We're working to protect our knowledge, our intellectual property under the UN's Nagoya Protocol. We don't want others patenting the guayusa root or leaf."

Beyond safeguarding knowledge, partners facilitated broader disseminations supporting training programs for farmers, co-creating regional agroforestry initiatives, and promoting regenerative practices through public events and networks. C1 collaborates with several stakeholders to promote regenerative practices, "Participate in the first summit of regenerative agriculture in Guatemala 'Guate Regenera'. The future of agriculture awaits you," (C1, social media).

4.1.6 | How Deep Ecological Knowledge Informs Market Engagement

Mobilizing ecological knowledge influences market engagement in three ways: first, by offering bundles of diverse products instead of single crops. Our cases design polycultural systems where multiple crops and livestock species interact to enhance pollination, nutrient cycling, pest control, and fertilization. Although ecologically advantageous, this bundle of products complicates commercialization in supply chains designed for single crops. The need to deal with this upstream diversity influences their downstream strategies in the marketplace.

Second, codification systems help translate ecological practices into credible quality signals that resonate with clients. By communicating biodiversity value to consumers, firms commercialize offerings that would otherwise remain undervalued or invisible. This way, respecting ecological principles becomes recognized and rewarded by the market.

Third, ensuring that RgSCs are both ecologically coherent and economically feasible requires firms to synchronize demand with supply. Biodiverse production cycles do not always align with conventional market dynamics, prompting firms to rethink product positioning, manage volumes, and balance temporalities across the supply chain. The market-oriented adjustments derived from mobilizing ecological knowledge constitute one side of the recursive relationship. The other side constitutes the second capability: reconfiguring market engagement, to which we now turn.

4.2 | Reconfiguring Market Engagement

This refers to the ability of firms to reshape how products, clients, and demand are managed so they respect socioecological boundaries and potentially align with socioecological cycles. Focal firms treat markets as flexible spaces in which firms can intentionally maneuver and even constrain market expectations so that consumption patterns do not exceed ecological boundaries but instead protect biodiversity and cultural specificity. Instead of market growth, firms cultivate demand, so it respects what can be produced within SES' limits. This capability is made of the following routines.

4.2.1 | Shaping Biodiverse Offerings

Shaping biodiverse offerings refers to the ability of firms to design a portfolio of products that add value and reflect the ecological diversity of their farming systems while remaining economically viable. To do so, all three cases strategically balance established products with emerging ones. The resulting combinations ensure economic viability while maintaining ecological commitments. As a codirector of C3 explained, "for now, we decided to focus on, let's say, the most commercial products. Of course the Chakra has many other small products which we are constantly testing in the market (...) but currently it is better to focus on these four products." This balancing process also accommodates species that mature at different rates: "while the agroforestry system of native species matures, we have other exotic species [not necessarily native] integrated as a way to generate income, for example, bananas and jackfruit," continued the codirector of C3.

Bundling allows lesser known or less popular species to "ride along" higher demand ones. As C2's cofounder noted: "Our most popular product is, by far, açai; our second one is the set that has açai with the rest of our products." This tactic goes further in C1 as explained by the cofounder: "The idea is not to just sell eggs, but all the animal proteins: cheese, chicken, meat. If a customer only comes to buy eggs, we don't sell."

Developing biodiversity-based markets takes time, especially when working with plants unknown to most customers. "We would like to accelerate the sale of Amazonian unconventional plants," said C2's cofounder, "but they don't have enough demand yet." Firms thus stagger introductions, building familiarity slowly. "It is a process that requires patience," continued the cofounder of C2. "You need time and capital to keep the rhythm of international business development." Our cases rely on a flagship species to "carry" the brand while creating demand for others. In C1, it is cheese and poultry; in C2, it is açai; and in C3, it is cocoa.

Even when markets favor specialization, firms resist monocropping. One of C3's directors explained: "We could focus only on cocoa or guayusa that are high in demand, but that is not how the Chakra works. Chakra states that we must have diversity." Firms also acknowledged the need for economic continuity in the face of seasonality. "Some products have limited harvesting periods, a season or two, so we explore other products to offer markets year-round," noted a codirector from C2.

Despite this intentional design, many species in the case studies remain valued only for their ecosystemic contributions (i.e., soil health, household diets, or minimal sales at local plazas), rather than as a source of revenue. A codirector of C3 elaborated: "They [referring to other products from the Chakra] are used for self-consumption, well sometimes a little gets sold at the local plaza, but it is minimum." In these cases, although ecological diversity is integral to production systems, the economic role of companion species is often secondary—enhancing the quality, resilience, or market appeal of the primary crop rather than a standalone revenue source.

Adding value at origin plays a central role in enabling firms to capture more value while balancing ecological integrity. To do so, they invest in local processing capacity, organizational infrastructure, and technologies that make biodiverse products more viable. A codirector from C3 explained: “We have already obtained our export license directly, and we have created our own export team and systems. This has made us stronger” C1 has taken a different route, bypassing supermarkets altogether in favor of direct sales through their own store; “in a regular supermarket,” explained the cofounder, “I would have to stand there with a megaphone explaining the product and its advantages in order for me to sell ... impossible. That is why we sell directly.”

C2 focuses on processing as a means of adding value at origin, particularly through freeze-drying, which preserves nutrients while improving logistics. A C2 cofounder explained: “It allows us to aggregate value and open access to international markets. Dehydration solves logistical issues in the Amazon. To extract a product 1000 kilometers inside the Amazon, it is much better to do it directly as a dry product. It is cheaper, lighter, and easier to transport.” Their vision, he continued, is to become a reference for dehydrated products from the Amazon, serving as a “bridge between the international market and the local territory.”

4.2.2 | Reframing Value

Reframing value refers to the routines through which focal firms challenge conventional market notions of worth, such as price or volume, by emphasizing ecological care, social contribution, and cultural origin, instead. Products are more than commodities; they tell a story that highlights their uniqueness, health benefits, and positive impact, reshaping what stakeholders consider valuable. This decommoditizing through quality, health, and place is key to steering markets toward regenerative goals. Without it, regenerative products would be forced to compete in commodity markets (i.e., coffee, cacao, or cheese) that reward extractive efficiency over socioecological care.

As the cofounder of C1 noted: “We have been teaching our customers that it is not just about consumption for consumption's sake, but about opting for quality.” Similarly, C3 refrains from selling sugar-heavy chocolate appealing to healthier attributes: “We promote healthy, uncontaminated food (...) aiming to contribute to societal health.”

Narratives also aim to elevate products from commodities to a connection to place, its people, and the broader ecosystems. A social media post from C1 captures this well: “Where does your food come from? Give life to your body with real food.” On C2's website, the connection to origin and health is even more explicit: “[C2's] purpose is to promote Amazon's biodiversity by enabling access for anyone on the planet to the nutritional, functional, and medicinal powers of plants.” The firm even presents their products as “medicine from the forest,” intertwining narratives of indigenous wisdom and biodiversity preservation in their marketing.

To further differentiate themselves, firms rely on developing trust through the cultivation of relationships and credibility. In all of our cases, trust is built through lived experiences: C1 employs a transparency-first strategy, fostering direct engagement through farm visits, public conferences, and consistent communication via social media. These practices allow consumers to witness the philosophy and operations behind regenerative agriculture firsthand. C3's codirector described how they welcome clients into the territory to experience production and their culture firsthand: “You can come visit in an in-person, ancestral immersion with our communities so you can see that the product we are offering is always organic.” The organization goes further by working with clients to offer this experience to customers: “We have worked together with our client in Japan to connect the consumer, via Zoom, to the Chakra (...). We have even received some customers from Japan. This year we received two groups of consumers that spent 10 days visiting several chakras and producers.”

Firms also rely on traceability and third-party certifications to generate credibility. This allows external validation while making the ecological and social practices behind each product more distinct. For C2, this includes several global certifications: “We are proud to share our global organic certifications for Europe, the United States, and Brazil (EC, USDA and BR), as well as the Halal Certificate ...” (webpage, C2).

By reframing value as relational and regenerative, firms are able to communicate, not only what a product is, but how it was made, where and by whom. This allows these firms to position themselves, and their buyers, as active participants in socioecological regeneration.

4.2.3 | Calibrating Demand and Supply

Calibrating demand and supply refer to the intentional effort to influence consumption and production in ways that respect socioecological limits. Calibrating helps ensure that regenerative efforts are not undermined by market pressures. Rather than allowing demand to drive production, firms attempt to recalibrate market expectations. They prioritize long-term ecosystem integrity over short-term volume gains, often by educating clients, selecting value-aligned partners, and pacing their growth.

Cases prioritize value-aligned clients and suppliers who understand and support their philosophy. C1, for example, restricts product availability to new customers, particularly during periods of high demand, such as December, when poultry or butter is in short supply. C1's cofounder noted: “We do not sell to new customers, to avoid shortages for regular and loyal customers.” Loyal customers benefit from greater consistency and reliability, knowing they will receive a steady supply of high-quality products, even in peak season. This selective approach extends to restaurants as well: Only those with close ties to the farm are served under a strict no-return, no-waste policy. As the cofounder noted, “There are many clients who want our product, and sometimes we don't have the product, sometimes it is out of season. So, if you can't manage

to educate them, it is normal that they end up leaving, but the ones who stay are loyal.” In essence, the firm fosters close relationships so clients adjust their purchases to what the farm has available, not the other way around.

This commitment to value-aligned relationships also shapes sourcing strategies. C2 grows both its own supply and also buys from nearby producers. Even if a contract has not yet materialized, the company maintains the relationship alive by exchanging knowledge, providing training, and buying products. The cofounder of C2 elaborated, “despite satisfying demand with our own supply, we sometimes buy from the communities near us (...) because we want to cultivate a relationship as we expect demand to grow.” This willingness to maintain supplier relationships ahead of demand reflects a long-term orientation and a logic of partnership rather than opportunism.

To avoid the extractive dynamics typical of conventional growth models, focal firms combine different forms of scaling: no growth, slow growth, and scaling outwards by replicating production across new communities rather than intensifying in existing farms. These choices are closely tied to ecological thresholds and biodiversity conservation.

C1 has deliberately set a ceiling on production, grounded in years of experimenting to finally reach an understanding of what their land can provide sustainably. “Our idea is to stay with a group of loyal customers. We have calculated that the farm has a capacity to produce animal protein for approximately 300 people per day—that is 150 grams per person. That is what we monitor: the limit of people we can sell to.” This reflects a steady-state logic where growth is not synonymous with success. Instead, the farm prioritizes its long-term balance, working with a loyal client base to ensure that production remains within its ecological means.

In C3, growth is intentionally slow. This approach was challenged when a major guayusa buyer requested a production boost, which the organization ultimately refused, due to biodiversity related concerns. Every supplier has a specific limit of production based on the Chakra system. A codirector explained: “Producers are contractually required not to expand beyond their registered agroforestry areas, this to avoid deforestation if we have more demand.” As such, growth is not driven by market pull but, by ecological feasibility, mapped and negotiated at the community level: “We assess how much area is available—two, three, five hectares—and analyze how that land is used for conservation, pasture, crops, household consumption. Based on the principles of Chakra, we map the total and define limits.” Field technicians play a key role in ensuring compliance, working closely with each farmer to uphold Chakra principles. When harvests are delivered, C3 checks whether the volumes fall within the expected range. If yields are significantly higher than estimated, they visit the farm to understand why. If for instance, deforestation is the cause, they issue sanctions and may eventually stop purchasing. Thus, instead of intensifying output, they slowly expanded production by integrating new suppliers but only after completing a certification process that ensures they comply with Chakra principles.

Meanwhile, C2 takes a more expansive view of growth. The founder's vision is to become a reference point for Amazon-sourced products, but growth is pursued through modular expansion rather than centralization. Only a portion of the firm's own farm is cultivated, whereas the rest is reserved for regeneration to grow and test more species. Rather than limiting demand, they intend to grow through replication. As previously explained in Section 4.1.3, C2 has managed to “pack” production transformation into a container/factory lab, which they expect to replicate in communities as they develop demand. The container allows closeness to production generating value-added activities that will make transportation viable but at the same time manage to capture more value closer to the place of origin. According to the cofounder, “in the future, yes, we can multiply and scale up this capacity in the Amazon Rainforest as it can be managed directly by groups of farmers. With this we can access different sources of ingredients using the container factory lab to bring these ingredients from the core of the Amazon rainforest to the main factory in Belém.”

Across our cases, growth is not intrinsically negative; it has to follow ecosystem's dynamics. As one codirector C3 expressed: “We do seek growth, but a patient one. We move like nature does: not fast, just slow.”

4.2.4 | Partnerships Enabling the Reconfiguration of Market Engagement

Partnerships also play an enabling role in reconfiguring market engagement, particularly because many firms lack capital and technical capacity needed to commercialize biodiverse products and navigate specialized markets. As with mobilizing ecological knowledge, external alliances help bridge these gaps. C3, which has limited market-oriented capabilities, illustrates this most clearly. To develop a viable market for guayusa, C3 partnered with a company interested in sourcing and commercializing guayusa-based products. This collaboration has supported value-adding activities at origin, including investments in drying and packaging machinery that are jointly owned by C3 and their partner. As one codirector explained, “this machine does not belong to [C3] ... actually a part of it belongs to us and the other belongs to our partner” Universities and mission-driven companies have also assisted C3 in designing and testing ready-to-eat products, such as cassava and plantain snacks, to strengthen the cooperative's capacity to sell directly to local consumers.

In contrast, C2, which has more developed commercial skills, focuses on partnerships with specialized retailers and microdistributors to expand its market reach. As explained in Section 4.2.3, C1 partners with several upscale restaurants that value regenerative sourcing and can communicate product distinctiveness to end consumers.

Across cases, partnerships complement internal capabilities by helping firms shape biodiverse offerings and reframe value through trusted intermediaries. These alliances therefore extend beyond transactional arrangements; they function as strategic relationships that enable regenerative products to gain legitimacy, reach appropriate markets, and sustain

the economic foundations required for long-term ecological stewardship.

4.2.5 | How Market Engagement Informs How to Mobilize Ecological Knowledge

This section examines the reverse side of the recursive relationship identified in Section 4.1.6 by showing how market engagement feeds back into the upstream mobilization of ecological knowledge. Engagement with downstream markets shapes what firms need to learn, how knowledge is applied, and which forms of knowledge must be formalized or shared across the supply chain.

First, market signals guide both the acquisition and application of ecological knowledge. Demand for certain species pushes firms to deepen their understanding of those species and their ecological dynamics, whereas weak market traction redirects attention toward alternatives with stronger commercial potential. A codirector of C3 explained how a price shock triggered this process: “In 2016, the price for cocoa declined heavily, which forced us to diversify and one year later, we incorporated plantain.”

Second, market expectations influence how ecological practices are implemented. Requirements from restaurants, retailers, and buyers shape adjustments in harvesting, process, and diversification strategies to achieve desired quality attributes and volumes. As demand for fine-flavor cacao has increased, a director of C3 described the resulting changes in production; “the idea with cacao is that we are focusing on improving production (...) Because we work with fine-flavor cacao, it sometimes needs more pruning or grafting so it can produce and be harvested more. We work with producers, explaining that yields can increase by applying certain practices depending on each plot (...) and with this, we expect production to increase next year.” Logistical constraints, including seasonality, shelf life, and volume requirements, further encourage firms to refine ecological practices so that supply aligns with the timing and structure of downstream markets. As explained in Section 4.1.3, investments in freeze-drying technology reflect the need to extend shelf life and increase volumes.

Third, market engagement strengthens the need for codification and knowledge exchange. Certifications, traceability demands, and quality standards require firms to document additional ecological practices or refine existing ones. As a result, C2 and C3 are in the process of obtaining EU deforestation-free and fair-trade certification. These cases are discussing with their suppliers the most efficient way to do this, taking advantage of previous certification processes. At the same time, concerns about misappropriation, such as the risk of external actors patenting guayusa, explained in Section 4.1.5, motivate codification for legal protection.

Together, these dynamics show how market engagement acts as a balancing pressure that refines and redirects ecological knowledge processes. Interacting with markets reveals which ecological attributes matter most, which practices require adjustment, and which forms of knowledge should be codified

and shared. Market engagement, therefore, is not only shaped by ecological learning but also actively shapes its evolution, reinforcing the recursive relationship at the core of regenerative supply chains.

5 | Discussion

Existing research on capabilities in sustainable SC (Pagell and Shevchenko 2014) largely conceptualizes them as responses to market pressures. As a result, capabilities are typically framed around incremental improvements, win-win solutions, and efficiency gains—assumptions that overlook the trade-offs inherent in supply chains that must sense, synchronize, and cocreate with living systems (Muñoz and Branzei 2021; Gualandris et al. 2024). Current sustainable SC scholarship therefore struggles to explain how firms can develop capabilities that operate within—rather than degrade—the SES that sustain them. This research advances an ecocentric understanding of supply chains by explaining how firms promote RgSCs through two regenerative capabilities and their recursive alignment, thereby moving beyond dominant sustainable SCM perspectives.

5.1 | Integrating Regenerative Capabilities

This research reveals two dynamic and interdependent capabilities—mobilizing deep ecological knowledge and reconfiguring market engagement—that enable focal firms to promote RgSCs by working within SES limits rather than responding solely to market signals. Each capability unfolds through its own recursive learning cycle. Mobilizing ecological knowledge unfolds through a cycle of acquiring, applying, codifying, and exchanging knowledge. Reconfiguring market engagement develops through shaping biodiverse offerings, reframing value, and calibrating demand and supply. Both cycles progress through experimentation and continuous feedback, as firms attune their decisions and operations to the dynamics of the ecosystems that sustain them. Through these processes, firms adjust their production systems considering ecological and market signals, making choices that frequently involve challenging trade-offs.

Although analytically separated, these capabilities function in practice as an integrated system. The model (Figure 2) introduces the notion of recursive alignment: Ecological knowledge informs market decisions, and market interactions prompt new learning about what is viable or worth adapting upstream. Sensemaking in complex ecological contexts is essential for organizational adaptation (Whiteman and Cooper 2011), and the cases show how firms embrace ecological complexity by acknowledging nature's agency and creating with—rather than controlling—the ecosystems they inhabit (Vlasov 2021).

Recursive alignment does not eliminate tensions between SES and commercial logics; it enables firms to confront them. Decisions about how much land to conserve, which species to prioritize, or when to privilege income generation are continuously renegotiated. Although all cases rely on cash crops to finance operations, they simultaneously cultivate lesser known species to enhance biodiversity and strengthen the resilience of their main crops. Climate variability, evolving ecological

knowledge, and shifting market feedback continually redefine what is viable or desirable.

The conceptual model empirically illuminates how focal firms shift from harm reduction toward regeneration. RgSCs must “restore a balanced variety of native organisms and species and adjust scale and scope of production and consumption within the proportions that SES can best tolerate” (Gualandris et al. 2024, 58). Firms cultivating these two capabilities develop a willingness to renounce certain market opportunities and navigate uncertainty with intention—going beyond incremental adjustments to enable more radical changes in how supply chains organize production and consumption.

This shift allows us to address a second major limitation in sustainable SCM literature: its reliance on engineering-based assumptions that treat supply chains as stable, controllable, and largely separable from the natural systems in which they operate. Sustainable SCM research has increasingly questioned theoretical approaches that assume stable operating conditions, predictable disruptions, and supply chains functioning independently of ecological systems. Such assumptions overlook that supply chains are embedded in dynamic SES whose conditions evolve across time, space, and meaning (Wieland 2021). As a result, existing theories struggle to explain how supply chains adapt, persist, or transform in the face of environmental change. Wieland (2021) therefore calls for reinterpreting supply chains through an ecological lens—one that foregrounds interdependence, adaptation, and relationality.

The conceptual model directly responds to this call by showing how regenerative focal firms develop capabilities that treat supply chains as ecologically embedded and relational systems. The capability of mobilizing ecological knowledge demonstrates how firms abandon assumptions of stability and instead cultivate routines for continuously interpreting ecological signals, assessing ecosystem conditions, and updating practices accordingly. These routines break with traditional ideas that supply chains can operate apart from the environments that sustain them.

Rather than seeking to return to a single optimal equilibrium, firms in this analysis accept uncertainty, embrace variability, and treat disturbances as learning opportunities. Their regenerative logic thereby contrasts sharply with the stability-oriented focus of dominant sustainable SCM frameworks.

The findings show that capabilities in regenerative contexts are not internally accumulated resources, but relational accomplishments built through interaction with nature, farmers, local communities, NGOs, researchers, buyers, and consumers. Unlike sustainable SCM approaches centered on monitoring and control (Erhun et al. 2021; Sodhi and Tang 2021), focal firms intentionally relinquish control in favor of deeper connection, prioritizing reciprocity, and ecological attunement over optimization.

A key insight is that capability development is grounded in place-based embeddedness. Firms' operations are physically located in the territories they regenerate, exposing them to constant ecological feedback and enabling “appropriate stewardship of both

the natural environment and other components” (Shrivastava and Kennelly 2013, 97). However, physical embeddedness alone is not unique to regenerative firms, as many vertically integrated corporations in agro-industrial sectors also operate in close proximity to their production zone. What differentiates firms in this research is not just geographic proximity but the relational and ecological awareness with which they inhabit these spaces.

This ecological embeddedness also reshapes market engagement. Firms shape biodiverse offerings, reframe value, and calibrate supply and demand through collaborative, relational strategies. They create markets for underrecognized species, decommoditize conventional crops, and educate buyers about ecological realities. Demand for regenerative products is thus cocreated rather than assumed (Reficco et al. 2018). Place-embeddedness enables firms to translate ecological constraints into credible and compelling market value propositions.

Finally, the conceptual model highlights a temporal logic that diverges from conventional supply chains' expectations. Regenerative firms adopt slowness, iteration, and ecological pacing, recognizing the biological cycles and cultural dynamics that shape production. Through recursive learning loops, firms operationalize the principle of proportionality (Gualandris et al. 2024), adjusting production and consumption to fit within SES limits. Value is thus redefined—not by volume or speed but by the integrity and alignment of processes embedded in place.

In sum, fostering regeneration in supply chains lies not only in combining ecological knowledge and market practices but in continuously realigning them with each other and with the socioecological context in which they are embedded. This alignment is relational, place-based, and recursive, offering a powerful counterpoint to dominant models of supply chain efficiency and growth.

5.2 | Reinterpreting the AAA Capabilities

Agility, adaptability, and alignment have long served as core capabilities in SCM, offering ways to respond to uncertainty and to build competitive advantage (Lee 2004). Yet these capabilities have also been largely defined through an economic lens—prioritizing, speed, responsiveness, and efficiency in serving markets, with limited attention to the environment and people (Erhun et al. 2021). Building on a conceptual model that reframes supply chains as embedded socioecological systems, we revisit the Triple-A framework. In doing so, we suggest how these capabilities can be extended to support long-term health of the SES of supply chains and economic performance.

In conventional supply chain thinking, agility refers to the ability to respond quickly to changes in demand (Lee 2004). When sustainability is introduced, agility expands to include responsiveness to a broader set of external stakeholders such as regulators, NGOs, and communities, who may trigger different sorts of risks (Sodhi and Tang 2021). The emphasis remains on speed and flexibility, but across more actors. This research suggests a different orientation: Agility is less about speed and more about rhythms. This aligns with the principle of polyrhythmicity (Gualandris et al. 2024), but these findings

provide a way to operationalize it through the mobilization of deep ecological knowledge. The cases align activities with ecological cycles (i.e., Chakra system), seasonal availability, and the temporalities of social systems (Rahman et al. 2024). Firms slow or sequence production to avoid excessive extraction and synchronize with multiple temporal rhythms (Muñoz and Cohen 2017). This suggests that agility in RgSCs is more about synchronicity than speed.

In traditional SCM, adaptability focuses on a firm's capacity to accommodate structural market shifts in supply or demand and spot trends in the marketplace (Lee 2004). In sustainability-oriented approaches, adaptability expands to develop innovative ways to control the supply chain and create market trends to make space for sustainable products (Erhun et al. 2021). However, focus is still inward-looking on “control,” centered on efficiency and responsiveness (Sodhi and Tang 2021). This research suggests that adaptability emerges through collaboration to respond to ecological boundaries. Firms codevelop offerings with suppliers, incorporating knowledge into decisions about what to grow, how, and when. Downstream, firms do not just create market trends but attempt to reframe value by developing relationships with clients to convey ecological knowledge. Clients themselves develop a certain degree of adaptability as they accept a limited number of products, buy in bundles, or not at all. Adaptability becomes a shared capability across actors in RgSCs.

Alignment refers to incentivizing partners to maximize performance. Sustainability concerns broadened this to include upstream and downstream actors, arguing that environmental and social factors often concentrate at the extreme ends of the supply chain (Sodhi and Tang 2021; Erhun et al. 2021). Yet alignment remains centered on minimizing risk and improving efficiency. It tends to focus on reducing harm rather than enabling regeneration, and it rarely accounts for the dynamic nature of the SESs that underpin production and supply.

This research suggests a more reciprocal and system-attuned form of alignment. Through ecological knowledge and market reconfiguration, alignment extends beyond transactional relationships to include ecosystems themselves. Examples include production ceilings based on Chakra principles (C3), livestock volumes tied to farm capacity (C1), and buying nonconventional plants despite no demand (C2). Nonhuman actors (soil, plant, culture, and community) gain agency and become participants in shaping constraints and possibilities. Recursive alignment also acts as a form of “reality check”: All the cases anchor their operation in cash crops to secure financial viability, with companion species supporting resilience rather than generating standalone revenue.

By reinterpreting the Triple-A framework through the lens of regenerative capabilities, this research advances understanding of what an ecocentric supply chain entails. When firms mobilize ecological knowledge, their practices extend beyond input substitution or stakeholder inclusion to cultivate capabilities that connect production knowledge with market engagement. In this reframing, agility is no longer defined by speed but by synchronicity with ecological rhythms; adaptability emerges as a shared relational capability across supply

chain actors; and alignment becomes a recursive, continuous process.

5.3 | Theoretical Contributions

This research makes four interrelated theoretical contributions to the supply chain management literature by advancing how capabilities are conceptualized in RgSC. First, we develop an empirically grounded model of RgSCs capabilities by identifying two recursive and interdependent capabilities—mobilizing deep ecological knowledge and reconfiguring market engagement—that enables firms to operate within socioecological limits rather than responding solely to market pressures. By foregrounding recursive learning cycles and ecological feedback, this contribution moves beyond dominant sustainable SCM perspectives that emphasize incremental efficiency improvements and harm reduction. Second, we introduce the concept of recursive alignment, extending prevailing understandings of alignment in supply chains by incorporating ongoing ecological and commercial feedback loops. Rather than threatening alignment as a static condition focused on incentive compatibility and efficiency, recursive alignment conceptualizes alignment as a continuous, relational process through which firms confront and renegotiate trade-offs between ecological integrity and market viability over time.

The third and fourth contributions extend existing theories by reinterpreting embeddedness and established supply chain frameworks through a regenerative lens. This research shows that RgSCs capabilities are grounded in place-based embeddedness paired with relational and ecological awareness. Whereas prior research acknowledges geographic proximity, our findings demonstrate that what differentiates regenerative firms is not location alone, but how firms inhabit place through sustained interaction with ecosystems and communities, thereby advancing embeddedness theory by explicitly linking it to capability development in SES. Finally, we extend the AAA framework (Lee 2004) by proposing regenerative interpretations of agility, adaptability, and alignment.

5.4 | Implications for Practice

This research offers several practical lessons for actors advancing RgSCs. First, organizations must expand their lens beyond conventional business metrics and financial returns to include the wellbeing of the SES in which supply chains are embedded. This means assessing success not only through productivity and revenue but also through soil health, biodiversity, livelihoods, and long-term landscape resilience. Practitioners should embed SES indicators into operational decisions and investment frameworks, recognizing that regenerative performance depends on long-term ecosystem functioning rather than short-term yield maximization.

Second, regenerative supply chains require a fundamental shift in how demand is understood and managed. Instead of treating consumer demand as fixed and limitless, RgSC actors must help shape consumption patterns around what ecosystems can sustainably produce. Anchor companies, buyers, and

consumer-facing brands have a role to play in communicating ecological constraints, transparency on trade-offs, and the additional social and ecological value created through regenerative practices. Educating consumers, adjusting market expectations, and aligning reward systems with ecological health are critical steps.

Policy and support organizations can accelerate regenerative transitions by enabling access to flexible credit, risk-sharing instruments, and incentives that reward stewardship across the whole supply ecosystem, not only commercial outputs. Supporting community-based actors, producer organizations, and local market linkages is essential to building fair, resilient, and regenerative value webs.

Finally, universities and extension systems should integrate regenerative thinking, system approaches, and indigenous and local knowledge into curricula and advisory services. Training future practitioners to work with diverse worldviews and ecological realities will help embed regenerative logics in practice, research, and policy. Notably, although our cases emerge in global south contexts where postgrowth narratives may be contested, these regenerative models align with principles of social-ecological wellbeing, careful out scaling, and systems thinking.

5.5 | Societal Implications

This research's ecocentric understanding of supply chains carry important societal implications. On the supply side, RgSCs fundamentally alter how production decisions are made by privileging SES boundaries rather than responding exclusively to market signals. Through the deployment of these capabilities, firms continuously adjust their production systems by integrating ecological and market feedback, confronting difficult trade-offs, and critically, developing a willingness to renounce certain market opportunities. This intentional navigation of uncertainty enables more radical changes in how supply chains organize production and consumption, moving beyond incremental efficiency gains toward practices aligned with ecological regeneration. On the demand side, RgSCs intervene in consumption patterns in ways that extend beyond the selection of certified products. Demand for regenerative products is not assumed but actively cocreated: Consumers are invited to adjust not only what they consume but also how much they consume, voluntarily aligning their consumption choices with ecological realities. In synthesis, RgSCs require both producers and consumers to acknowledge nature's agency and to create with—rather than seek to control—the ecosystems they inhabit. Societally, this implies a shift toward supply–demand relations grounded in interdependence, adaptation, and relationality, where actors cultivate tolerance for uncertainty, intentionally relinquish control, and prioritize reciprocity and ecological attunement over optimization.

5.6 | Limitations and Further Research

Although our study offers a foundation for understanding RgSCs, it also entails limitations that point to several avenues for future research. Our empirical focus on agribusiness firms

occupying similar positions within RgSC enables analytical depth but limits the generalizability of the findings across other supply chain roles and sectors. Because the theoretical contributions emphasize place-based embeddedness and recursive alignment with SES, future research should examine whether similar capabilities can emerge in downstream firms that are more distant from production territories and have weaker direct ties to ecosystems. A relevant open question is whether governance mechanisms, intermediaries, or partnerships can substitute for or complement territorial embeddedness. Addressing this would clarify whether territorial rootedness is a contingent enabler or a necessary condition for regenerative transformation and whether the capabilities identified extend beyond agribusiness to industries with different resources dependencies and ecological impacts.

Relatedly, the regional focus on Latin America allowed us to theorize RgSCs in contexts characterized by high biodiversity, social inequality, and institutional heterogeneity. However, this limits the transferability of our insights to other ecological and institutional settings. Comparative research across regions—such as tropical versus temperate ecosystems or developed versus emerging economies—could enrich understanding of how institutional arrangements, ecological conditions, and market structures shape regenerative capabilities and recursive alignment. Such studies could help identify context-specific pathways to regeneration.

Our findings also open a research agenda on growth and scaling in RgSCs. As shown, RgSCs combine different growth trajectories—no growth, slow growth, and growth through replication—by explicitly linking expansion decisions to ecological thresholds and biodiversity conservation. Future research could examine how these alternative growth models relate to the principle of proportionality (Gualandris et al. 2024) and whether constraining growth or pursuing replication-based scaling enhances socioecological balance. Advancing this line of inquiry requires research designs capable of testing cause-and-effect relationships using ecological and biological data, as well as closer collaboration between management scholars and ecological scientists. Such work is essential to move regeneration research beyond normative claims and assess whether the capabilities identified here actually contribute to the restoration of SES. In addition, further research should examine the economic sustainability of RgSCs that limit growth or expand through scale-out strategies, evaluating whether networked replication represents a viable scaling logic across sectors and supply chain configurations.

Finally, our findings raise broader questions about diversity, redundancy, and economic organization in RgSCs. By prioritizing biodiversity, RgSC departs from monocultural efficiency and economies of scale, instead emphasizing economies of scope, place-fit, and resilience. Although this research theorizes these features as central to regenerative capability development, future research should systematically examine the trade-offs between diversity, redundancy, and economic efficiency. Particularly, scholars could investigate the extent to which producers can sustain production diversity when not all outputs find commercial markets and whether economic incentives are sufficient to maintain diversity over time. Additional research is also needed

to understand how regenerative value is preserved as products move through intermediaries and if closer vertical integration or alternative governance arrangements are required. Addressing these questions would benefit from comparative research designs that assess ecological and economic performance before and after environmental or market shocks, allowing scholars to test whether diversity and redundancy indeed strengthen the resilience of SES. Extending this work across sectors would further clarify if economies of scope constitute a dominant economic logic in RgSCs.

6 | Conclusion

As humanity approaches critical ecological tipping points, the need to fundamentally rethink production and consumption systems has never been more urgent. In this context, RgSCs emerge not merely as a promising innovation but as a necessity for operating within the Earth's carrying capacity. This research shows how focal firms in Latin American RgSCs are advancing this transition by developing capabilities that reconnect production and consumption with the dynamics of living systems. By mobilizing ecological knowledge and reconfiguring market alignment, these firms operate within ecological thresholds while sustaining viable livelihoods, illustrating that regeneration is a process of continual adaptation rather than an attainable end state.

Interpreting our findings through the Triple-A framework extends this foundational perspective of supply chain capabilities into an ecocentric paradigm. RgSCs demand agility not only to respond to markets or stakeholders but also to align with multiple temporal rhythms—especially those of SES. Adaptability extends beyond innovating control on supply chains or creating market trends; it entails collaborating across the supply chain to respond to ecological constraints. Finally, alignment becomes recursive, reaching beyond efficiency-driven coordination to integrate ecological and commercial feedback loops that account for the SES dynamics underpinning production and trade. In doing so, RgSCs challenge the anthropocentric assumptions that have long guided supply chain design, operation, and governance.

Building on the theoretical contributions, identifying regenerative capabilities, introducing the concept of recursive alignment, foregrounding place-based embeddedness, and extending the AAA framework through ecocentric lenses, our findings also offer a clear practical insight: Regenerative transformation cannot be achieved through incremental improvements to conventional models. It requires a reorientation of core business assumptions. RgSCs prioritize the health and recovery of the SES in which they operate, treating natural and social capital as essential assets to be regenerated. They grow through scale-out rather than scale-up models—expanding through replication and collaboration instead of centralization and relying on bottom-up rather than top-down governance. They favor differentiation over standardization and commoditization, and diversity and redundancy over the lean-efficiency orthodoxy that treats redundancy as waste. Finally, they pursue economies of scope rather than economies of scale, integrating multiple products, services, and relationships to reinforce ecological and social

regeneration. In doing so, regenerative firms articulate a new narrative for economic activity—one that recognizes, as Keat Raworth aptly observes, that “growth is a healthy phase of life, but nothing grows forever.”

As the climate crisis deepens and biodiversity loss accelerates, the question is no longer whether regenerative approaches are necessary—but whether we can transition in time. As ecological movements remind us: “We are not defending nature. We are nature defending itself.” The future of SCM may well depend on how effectively we respond to this call and reorient our systems toward the regeneration of life itself.

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Conflicts of Interest

The authors declare no conflicts of interest.

Endnotes

¹The case study protocol is available online at the following repository: https://drive.google.com/drive/folders/18ozGJKg07TFnfwk8__OuDuJovVNHNVcH?usp=sharing.

²Syntropic agriculture is a movement with theoretical and practical implications for production systems in which the natural processes are translated into farming interventions (Götsch 1995).

³The chakra is formally defined by a 2017 municipal ordinance as a family-managed, biodiverse production space that integrates food, timber, medicinal, and other species while valuing ancestral agro-ecological practices and avoiding monocultures (Gobierno Autónomo Descentralizado Provincial de Napo 2017).

References

- Amir, S., N. Salehi, M. Roci, S. Sweet, and A. Rashid. 2023. “Towards Circular Economy: A Guiding Framework for Circular Supply Chain Implementation.” *Business Strategy and the Environment* 32, no. 6: 2684–2701.
- Bag, S., A. Chiarini, and G. Srivastava. 2025. “Building Regenerative Supply Chains: A Qualitative Study of SMEs in the Agro-Food Chains.” *Business Strategy and the Environment* 34: 10803–10823.
- Barney, J. 1991. “Firm Resources and Sustained Competitive Advantage.” *Journal of Management* 17, no. 1: 99–120.
- Creswell, J. W. 2013. *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*. SAGE Publications.
- Ellen MacArthur Foundation. n.d. “What is the Meaning of a Circular Economy and What Are the Main Principles? Accessed October 2023. <https://www.ellenmacarthurfoundation.org/topics/circular-economy-introduction/overview>.
- Erhun, F., T. Kraft, and S. Wijnsma. 2021. “Sustainable Triple-A Supply Chains.” *Production and Operations Management* 30, no. 3: 644–655.
- Eskandarpour, M., P. Dejax, J. Miemczyk, and O. Péton. 2015. “Sustainable Supply Chain Network Design: An Optimization-Oriented Review.” *Omega* 54: 11–32.
- Felin, T., N. J. Foss, and R. E. Ployhart. 2015. “The Microfoundations Movement in Strategy and Organization Theory.” *Academy of Management Annals* 9, no. 1: 575–632.

- Fisher, M. L. 1997. "What is the Right Supply Chain for Your Product?" *Harvard Business Review* 75: 105–117.
- Friedman, M. 1951. "Neo-Liberalism and Its Prospects." *Farmand*: 89–93. https://miltonfriedman.hoover.org/friedman_images/Collections/2016c21/Farmand_02_17_1951.pdf.
- Gereffi, G. 1994. "The Organization of Buyer-Driven Global Commodity Chains: How US Retailers Shape Overseas Production Networks." In *Commodity Chains and Global Capitalism*, edited by G. Gereffi and M. Korzeniewicz, 95–122. Praeger.
- Gioia, D. A., K. G. Corley, and A. L. Hamilton. 2013. "Seeking Qualitative Rigor in Inductive Research: Notes on the Gioia Methodology." *Organizational Research Methods* 16, no. 1: 15–31.
- Glaser, B., and A. Strauss. 2017. *Discovery of Grounded Theory: Strategies for Qualitative Research*. Routledge.
- Gobierno Autónomo Descentralizado Provincial de Napo. 2017. "Ordenanza que Regula y Reconoce el Sistema Productivo Chakra en la Provincia de Napo." Napo, Ecuador.
- Götsch, E. 1995. *Break-Through in AGRICULTURE*, 22. AS-PTA.
- Gualandris, J., O. Branzei, M. Wilhelm, et al. 2024. "Unchaining Supply Chains: Transformative Leaps Toward Regenerating Social-Ecological Systems." *Journal of Supply Chain Management* 60, no. 1: 53–67.
- Gunderson, L. H., and C. S. Holling. 2002. *Panarchy: Understanding Transformations in Human and Natural Systems*. Island Press.
- Hahn, T., and M. Tampe. 2021. "Strategies for Regenerative Business." *Strategic Organization* 19, no. 3: 456–477.
- Heikkurinen, P., J. Rinkinen, T. Järvensivu, K. Wilén, and T. Ruuska. 2016. "Organising in the Anthropocene: An Ontological Outline for Ecocentric Theorising." *Journal of Cleaner Production* 113: 705–714. <https://doi.org/10.1016/j.jclepro.2015.12.016>.
- Hellweg, S., E. Benetto, M. A. J. Huijbregts, et al. 2023. "Life-Cycle Assessment to Guide Solutions for the Triple Planetary Crisis." *Nature Reviews Earth and Environment* 4: 471–486.
- Hong, J., Y. Zhang, and M. Ding. 2018. "Sustainable Supply Chain Management Practices, Supply Chain Dynamic Capabilities, and Enterprise Performance." *Journal of Cleaner Production* 172: 3508–3519.
- IPBES. 2019. In *Global Assessment Report on Biodiversity and Ecosystem Services of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services*, edited by J. Brondizio, S. Settele, S. Díaz, and H. T. Ngo. IPBES Secretariat. 1148 pages. <https://doi.org/10.5281/zenodo.3831673>.
- Lee, H. L. 2004. "The Triple-A Supply Chain." *Harvard Business Review* 82, no. 10: 102–113.
- Luzzini, D., M. Pagell, V. Devenin, J. Miemczyk, A. Longoni, and B. Banerjee. 2024. "Rethinking Supply Chain Management in a Post-Growth Era." *Journal of Supply Chain Management* 60, no. 4: 92–106.
- Magnani, G., and D. Gioia. 2023. "Using the Gioia Methodology in International Business and Entrepreneurship Research." *International Business Review* 32, no. 2: 102097.
- Maxwell, J. A. 2013. *Qualitative Research Design: An Interactive Approach*. 3rd ed. SAGE Publications.
- Montabon, F., M. Pagell, and Z. Wu. 2016. "Making Sustainability Sustainable." *Journal of Supply Chain Management* 52, no. 2: 11–27.
- Muñoz, P., and O. Branzei. 2021. "Regenerative Organizations: Introduction to the Special Issue." *Organization & Environment* 34, no. 4: 507–516.
- Muñoz, P., and B. Cohen. 2017. "Towards a Social-Ecological Understanding of Sustainable Venturing." *Journal of Business Venturing Insights* 7: 1–8.
- Pagell, M., and A. Shevchenko. 2014. "Why Research in Sustainable Supply Chain Management Should Have No Future." *Journal of Supply Chain Management* 50, no. 1: 44–55.
- Pagell, M., and Z. Wu. 2009. "Building a More Complete Theory of Sustainable Supply Chain Management Using Case Studies of 10 Exemplars." *Journal of Supply Chain Management* 45, no. 2: 37–56.
- Parmigiani, A., and J. Howard-Grenville. 2011. "Routines Revisited: Exploring the Capabilities and Practice Perspectives." *Academy of Management Annals* 5, no. 1: 413–453.
- Patton, M. Q. 2002. *Qualitative Research & Evaluation Methods*. 3rd ed. SAGE Publications.
- Peng, D. X., R. G. Schroeder, and R. Shah. 2008. "Linking Routines to Operations Capabilities: A New Perspective." *Journal of Operations Management* 26, no. 6: 730–748.
- Porter, M. E. 1989. "How Competitive Forces Shape Strategy." In *Readings in Strategic Management*, edited by D. Asch and C. Bowman, 133–143. Palgrave. https://doi.org/10.1007/978-1-349-20317-8_10.
- Rahman, S., N. T. Nguyen, and N. Slawinski. 2024. "Regenerating Place: Highlighting the Role of Ecological Knowledge." *Organization & Environment* 37, no. 3: 466–494.
- Reficco, E., R. Gutiérrez, M. H. Jaén, and N. Auletta. 2018. "Collaboration Mechanisms for Sustainable Innovation." *Journal of Cleaner Production* 203: 1170–1186.
- Seuring, S., S. Aman, B. D. Hettiarachchi, F. A. de Lima, L. Schilling, and J. I. Sudusinghe. 2022. "Reflecting on Theory Development in Sustainable Supply Chain Management." *Cleaner Logistics and Supply Chain* 3: 100016.
- Shrivastava, P., and J. J. Kennelly. 2013. "Sustainability and Place-Based Enterprise." *Organization & Environment* 26, no. 1: 83–101.
- Sodhi, M. S., and C. S. Tang. 2021. "Extending AAA Capabilities to Meet PPP Goals in Supply Chains." *Production and Operations Management* 30, no. 3: 625–632.
- Srivastava, S. K. 2007. "Green Supply-Chain Management: A State-Of-The-Art Literature Review." *International Journal of Management Reviews* 9, no. 1: 53–80.
- Strauss, A., and J. Corbin. 1994. *Grounded Theory Methodology: An Overview*. Sage Publications Inc.
- Strauss, A., and J. Corbin. 1998. *Basics of Qualitative Research: Techniques and Procedures for Developing Grounded Theory*. 2nd ed. SAGE Publications.
- Stuart, I., D. McCutcheon, R. Handfield, R. McLachlin, and D. Samson. 2002. "Effective Case Research in Operations Management: A Process Perspective." *Journal of Operations Management* 20, no. 5: 419–433.
- Suárez-Eiroa, B., E. Fernández, G. Méndez-Martínez, and D. Soto-Oñate. 2019. "Operational Principles of Circular Economy for Sustainable Development: Linking Theory and Practice." *Journal of Cleaner Production* 214: 952–961.
- Teece, D. J. 2007. "Explicating Dynamic Capabilities: The Nature and Microfoundations of (Sustainable) Enterprise Performance." *Strategic Management Journal* 28, no. 13: 1319–1350.
- Teece, D. J., G. Pisano, and A. Shuen. 1997. "Dynamic Capabilities and Strategic Management." *Strategic Management Journal* 18, no. 7: 509–533.
- Touboulis, A., L. McCarthy, and L. Matthews. 2020. "Re-Imagining Supply Chain Challenges Through Critical Engaged Research." *Journal of Supply Chain Management* 56, no. 2: 36–51.
- Vlasov, M. 2021. "In Transition Toward the Ecocentric Entrepreneurship Nexus: How Nature Helps Entrepreneurs Make Ventures More Regenerative Over Time." *Organization & Environment* 34, no. 4: 559–580.

- Whiteman, G., and W. H. Cooper. 2011. "Ecological Sensemaking." *Academy of Management Journal* 54, no. 5: 889–911.
- Wieland, A. 2021. "Dancing the Supply Chain: Toward Transformative Supply Chain Management." *Journal of Supply Chain Management* 57, no. 1: 58–73.
- Winter, S. G. 2003. "Understanding Dynamic Capabilities." *Strategic Management Journal* 24, no. 10: 991–995.
- Yin, R. K. 2015. "Qualitative Research From Start to Finish." In *Research Methods*, 2nd ed. Guilford Press.
- Zhang, A., J. X. Wang, M. Farooque, Y. Wang, and T. M. Choi. 2021. "Multi-Dimensional Circular Supply Chain Management: A Comparative Review of the State-of-the-Art Practices and Research." *Transportation Research Part E: Logistics and Transportation Review* 155: 102509.

Appendix A

Sources of Information

	(C1)	(C2)	(C3)
Field visit	July 2023	July 2023	August 2023
Informant Job title	4 interviews: • Cofounder and current owner 1 (2 interviews) • Cofounder and current owner 2 (2 interviews)	6 interviews: • Founder (2 interviews) • COO • Market analyst • Product developer • Representative of Earth Impact (main investors)	11 interviews: • Cooperative director and general coordinator • Cofounder of cooperative (2 interviews) • Cooperative director and responsible of sales (2) • Cooperative director and responsible of organic certification • Cooperative worker responsible of training field technicians • Cooperative worker and responsible of organic certification • Cooperative worker and field technician 1 and 2 • Cooperative worker and responsible of supervision and formation of suppliers 1 and 2
Follow-up discussions	Discussion with lead researchers October 2023 November 2023 December 2023	Discussion with lead researchers October 2023 November 2023 December 2023	Discussion with other researchers October 2023 November 2023 December 2023
Interview updates	Interview with CEO August 2025	Interview with CEO August 2025	Interview with a director August 2025
Main secondary sources	One internal document related to strategic and financial planning https://www.facebook.com/LaGranjadeLolaGT/ https://www.instagram.com/granja.de.lola/ https://www.instagram.com/elmercaditodelola/?hl=en https://www.youtube.com/@granjadelolagt	Three internal documents related to strategic planning, sourcing and scaling provided by the company https://hortadaterra.com.br/ https://www.youtube.com/@hortadaterra9588 https://www.instagram.com/hortadaterra/	One internal document related to strategic planning FAO. (2023). Sistematización de las experiencias y procesos del Proyecto Conservación y Buen Vivir GEF Napo. Chancosa, C (2023). Actualización del plan estratégico con enfoque de sostenibilidad de la asociación Agro Artesanal Wiñak, en la provincia de Napo, Ecuador. Tesis de Maestría. Centro Agronómico Tropical de Investigación y Enseñanza (CATIE), Costa Rica. Lopez, T. (2022) “Modelo de gestión administrativo financiero para la Asociación Agro Artesanal Wiñak, del cantón Archidona, provincia de Napo, período 2021”. Tesis de Licenciatura en Finanzas, Escuela Superior Politécnica de Chimborazo. Ecuador https://www.winak.org/ https://www.corporacionChakra.org/asociacion-winak/
Example of other secondary sources	https://futuroverde.org/2020/10/el-mercadito-de-lola-una-granja-regenerativa/	https://www.linkedin.com/company/horta-da-terra/ https://www.facebook.com/HortadaTerra/?_rdc=1&_rdr https://pcabhub.org/en-us/news/news-highlights/fund-designed-in-partnership-with-usaid-brazil-invests-in-businesses-with-positive-impact-in-the-amazon	https://www.youtube.com/@destinationdevelopment8753 https://www.fao.org/giahs/giahsaroundtheworld/designated-sites/latin-america-and-the-caribbean/amazon-chakra/es/ https://www.lasillavacia.com/silla-academica/la-chagra-amazonica-ensena-como-producir-comida-sin-acabar-la-selva/ https://www.bivica.org/files/5771_Modelo%20Grupo%20Chakra_14.12.2020.pdf

Source: Created by the authors.